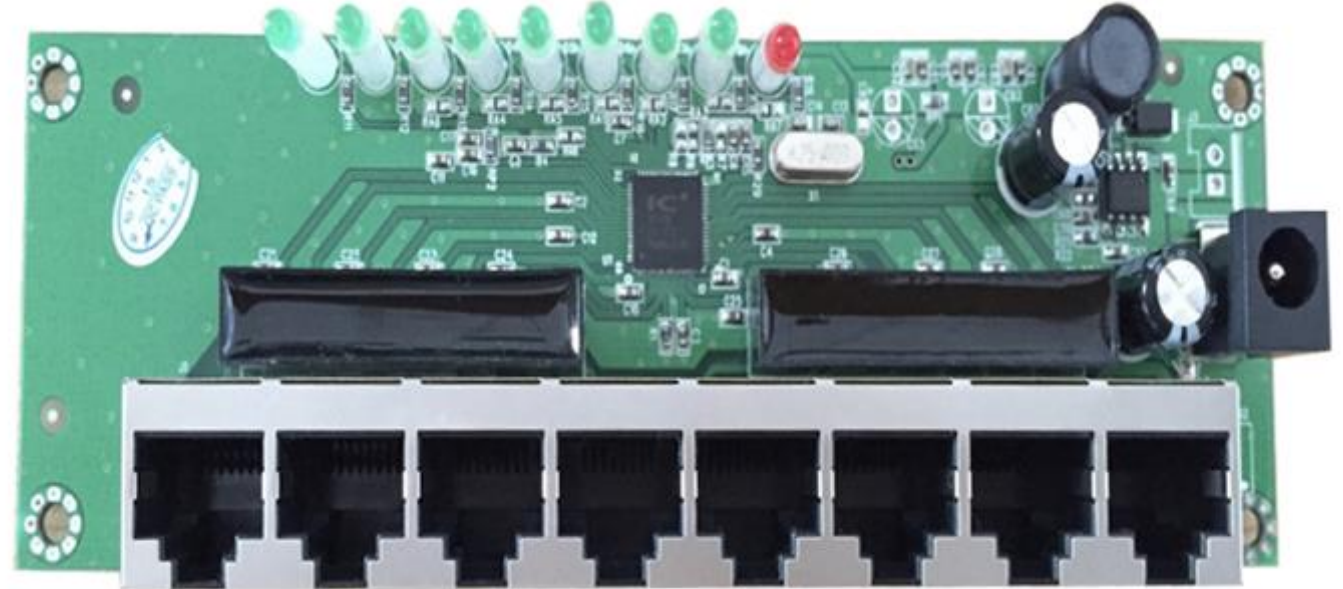


CS 31204: Computer Networks – The Routers

Department of Computer Science
and Engineering



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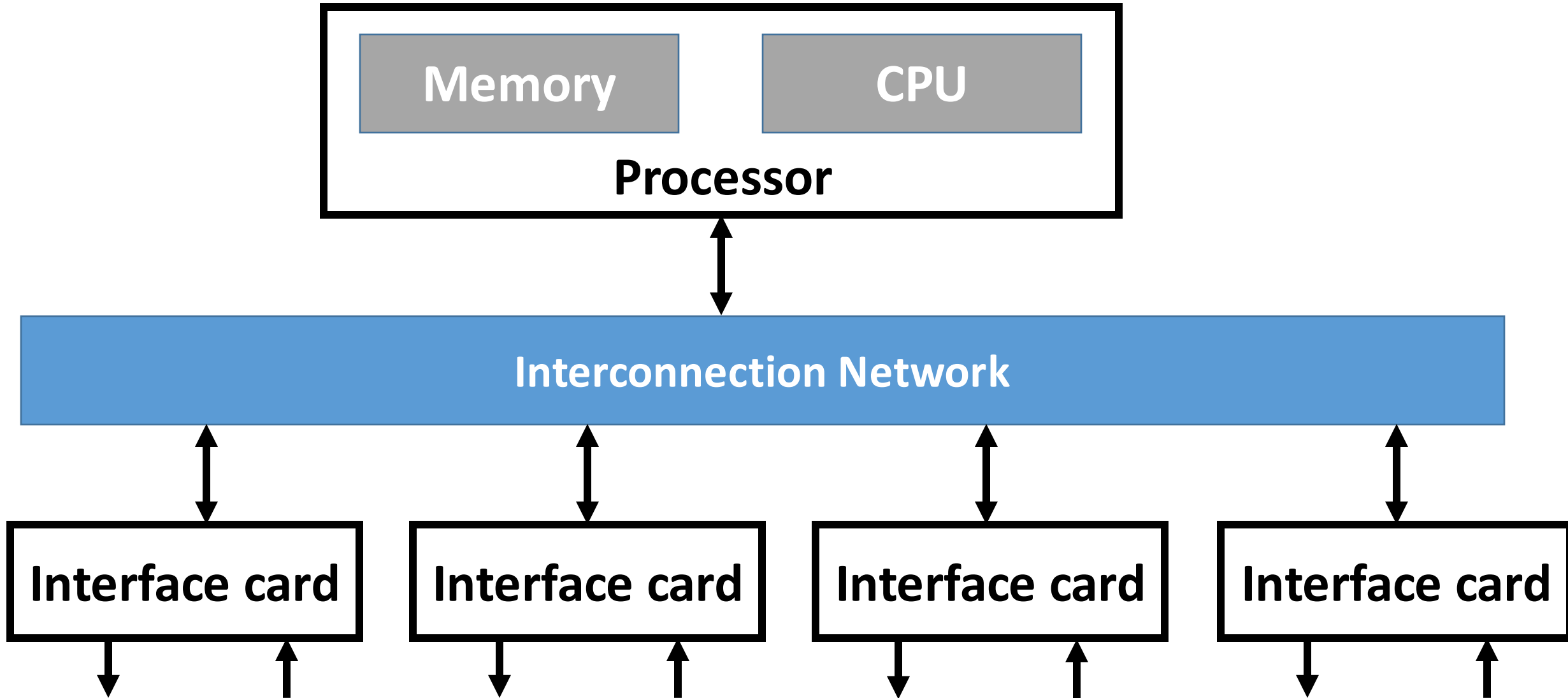
The Life of a Router

- Do
 - Find Path
 - Forward, forward, forward, forward, forward
 - Find Path
 - Forward, forward, forward, forward, forward
- Repeat until powered off
- Two basic operations –
 - Construct the routing table – **the control plane**
 - Do a routing match and forward the packet to a dedicated interface – **the data plane**

Evolution of Router Architecture

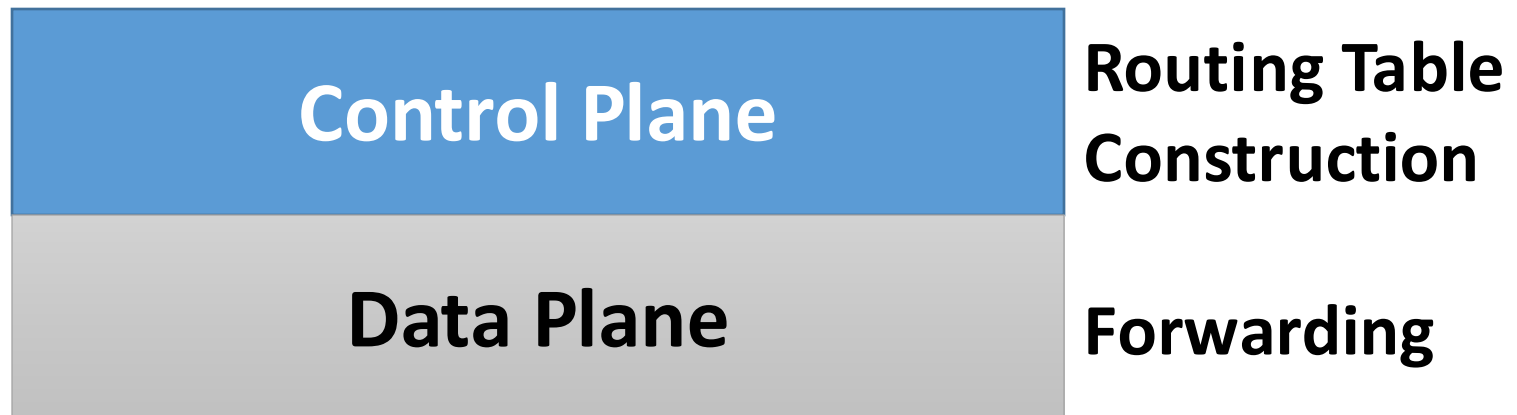
- 1st Generation (Until 1980's): Standard Computer
- 2nd Generation (Early 1990's): Delegate to interfaces
- 3rd Generation (Late 1990's): Distributed Architecture
- 4th Generation (Early 2000's): Distributed over multiple racks
- 5th Generation (Today): Software Defined Routing

Basic Architectural Components of a Router

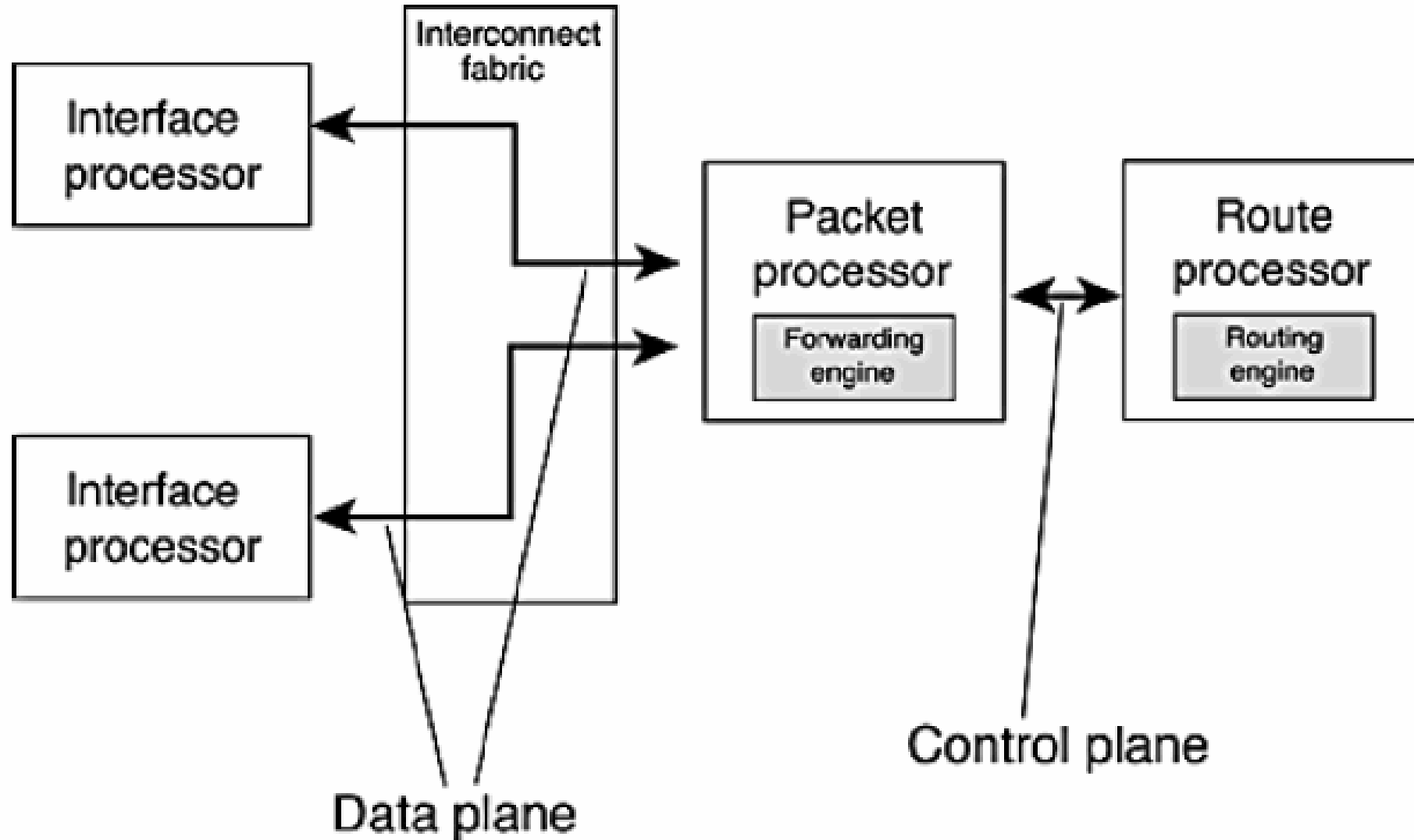


Router Hardware

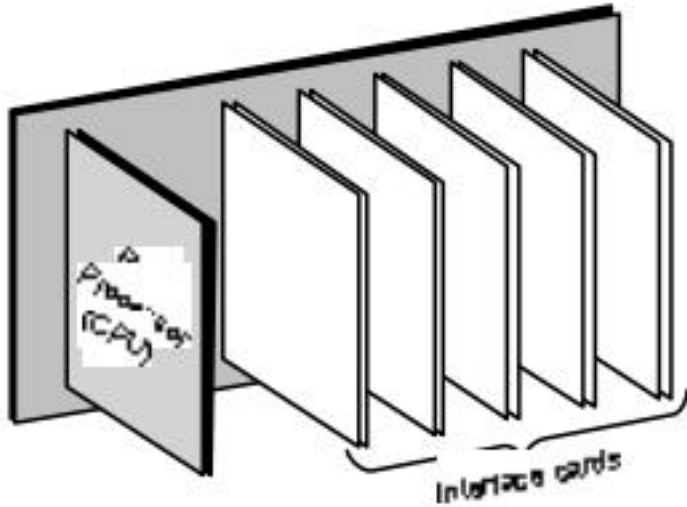
- Processor is responsible for control functions (**route processors**)
 - Construct the routing table based on the routing algorithm
- Forwarding is done at the interface card
 - Route match needs to be very fast
 - Specialized hardware – Ternary Content-Addressable Memory (TCAM)



Router Internals

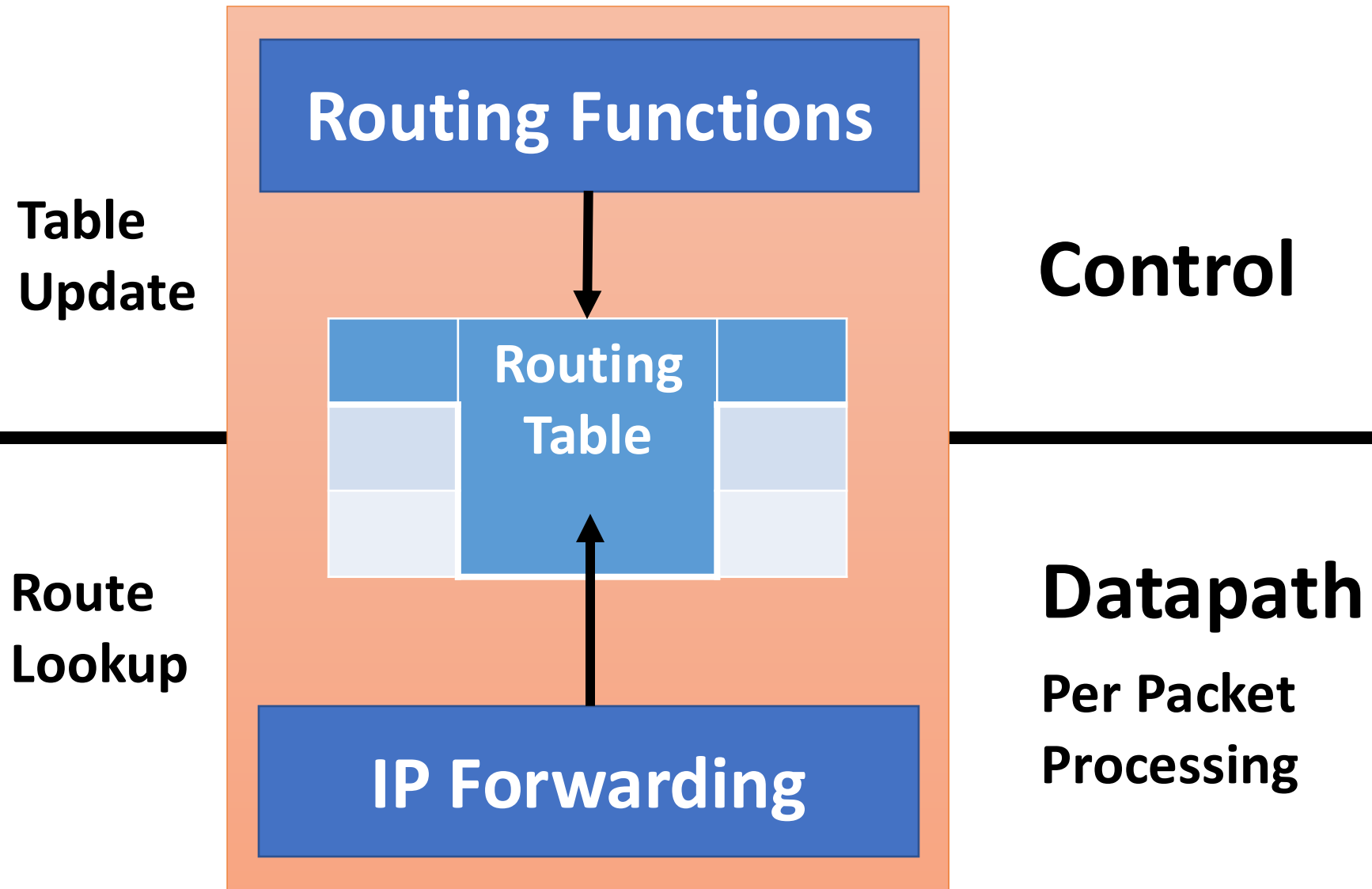


Slotted Chassis – The Physical Infrastructure for Routers



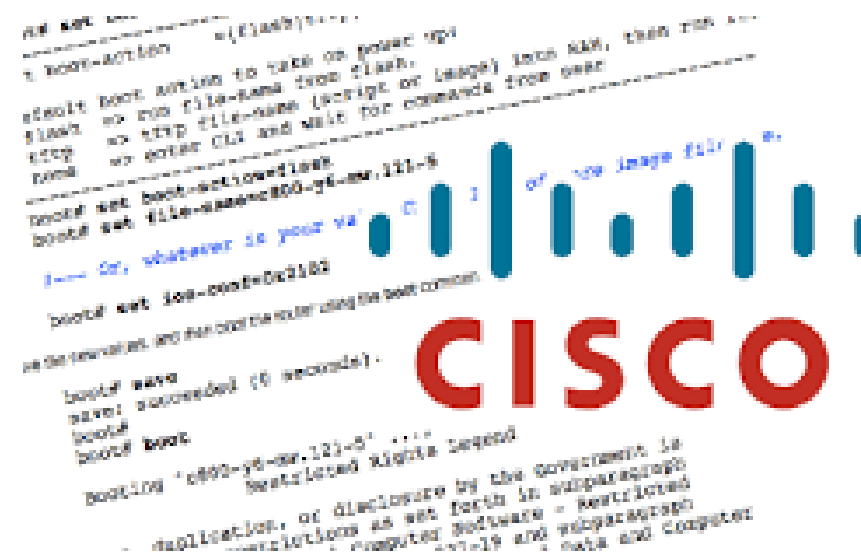
- Large routers are built as slotted chassis
 - Interface cards are inserted in the slots
 - Route processor is also inserted at a slot
- Simplifies repairs and upgrades of components

Functional Components



Control Plane in a Router

- Note that a router is a **special purpose computer**
- Implemented as a software (router OS) that supports the basic computing functionalities to run a router along with routing functionalities
- Routing protocols are implemented in the router OS
- Example: Cisco IOS



Routing Functions

- **Route Calculation**
- **Maintenance of the routing table**
- **Execution of the routing protocol**

- On commercial routers, routing functions are handled by a single general purpose processor, called the **route processor**

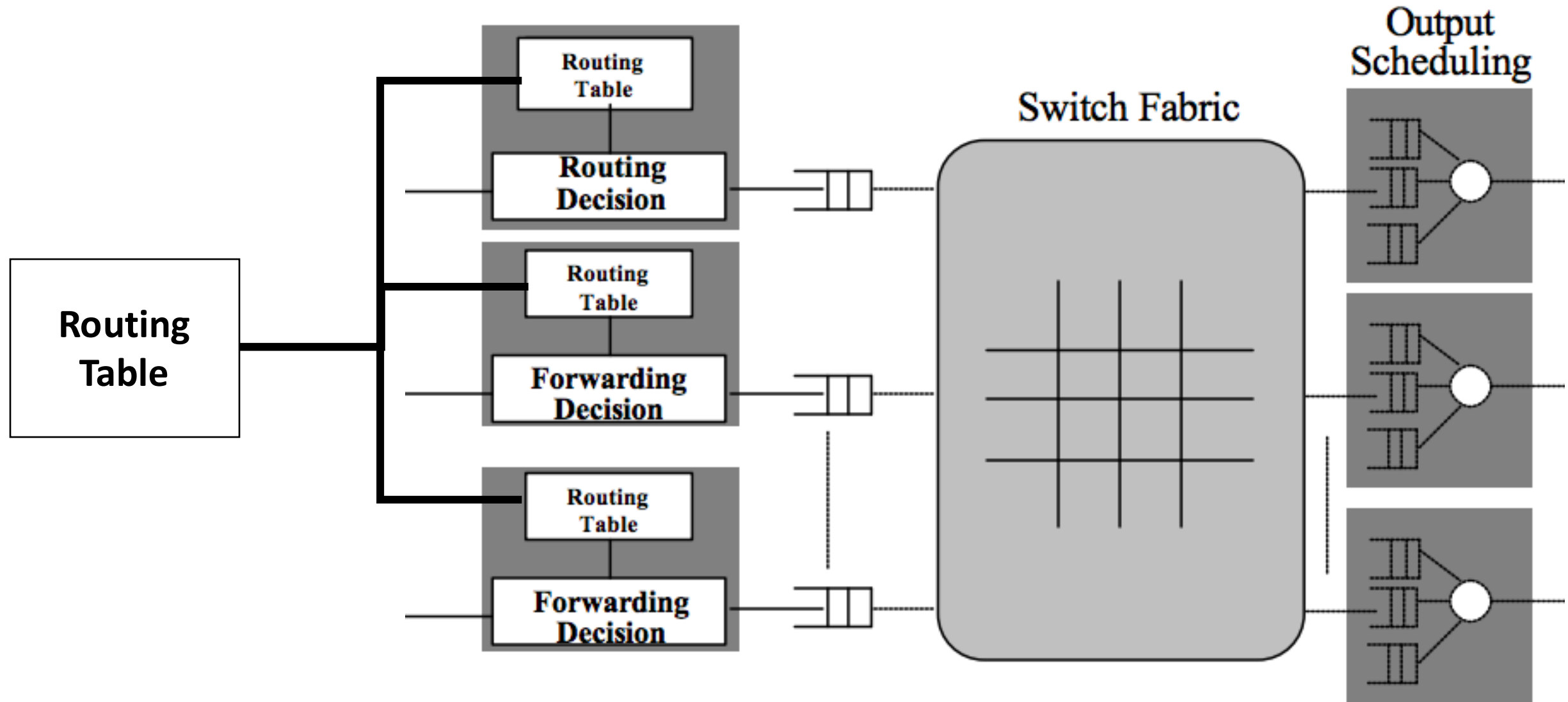
Data Plane of a Router

- Implement forwarding functionalities – make a route lookup and forward the packet at the destination interface
- Functionality is similar to a L2 switch – use switch fabric (the mapping from input ports to output ports) to forward the packet from one interface to another
- Maintains interface buffer – to implement store and forward functionality

IP Forwarding

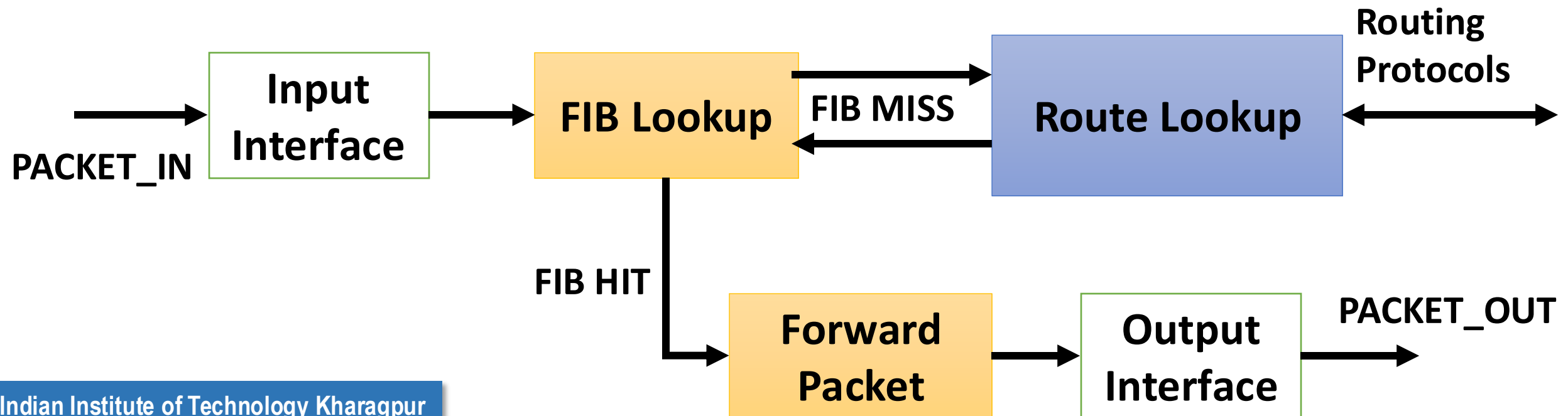
- Per packet processing of the IP packets
- IP forwarding is distributed, handled by individual interface controllers
- Special hardware devices are used - TCAM

Per Packet Processing – Basic Architectural Components



Forwarding Information Base (FIB)

- The interfaces maintains a *forwarding information base* (FIB) – a mapping from input interface to output interface
- A replica of the routing table used at the interfaces for making the forwarding decision



Difference between RIB and FIB

- **Routing Information Base (RIB)** – The routing table, implemented in software, is maintained at the control plane
- **Forwarding Information Base (FIB)** – The copy of the required routes maintained in interface TCAM hardware
- RIB is dynamic and maintains entire routing information, FIB is updated whenever required

RIB and FIB

The RIB

172.16.1.0	255.255.255.0	172.16.1.2	Eth0
172.16.2.0	255.255.255.0	172.16.2.2	Eth1
10.3.0.0	255.255.0.0	10.3.1.1	Eth3
10.9.0.0	255.255.0.0	10.9.1.1	Eth4



172.16.2.0	255.255.255.0	172.16.2.2	Eth1
10.3.0.0	255.255.0.0	10.3.1.1	Eth3
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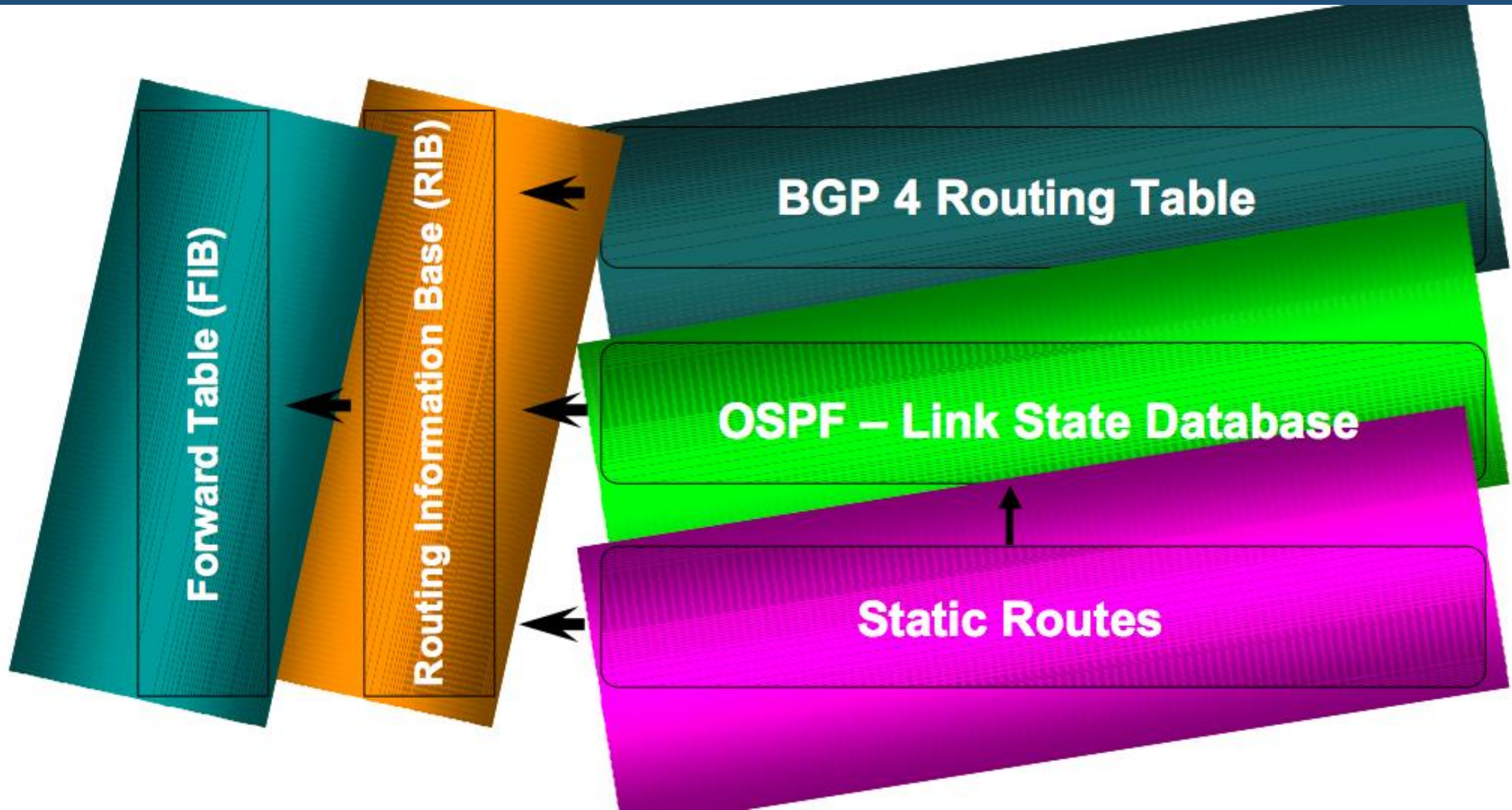
FIB at Eth0

172.16.1.0	255.255.255.0	172.16.1.2	Eth0
10.3.0.0	255.255.0.0	10.3.1.1	Eth3



FIB at Eth1

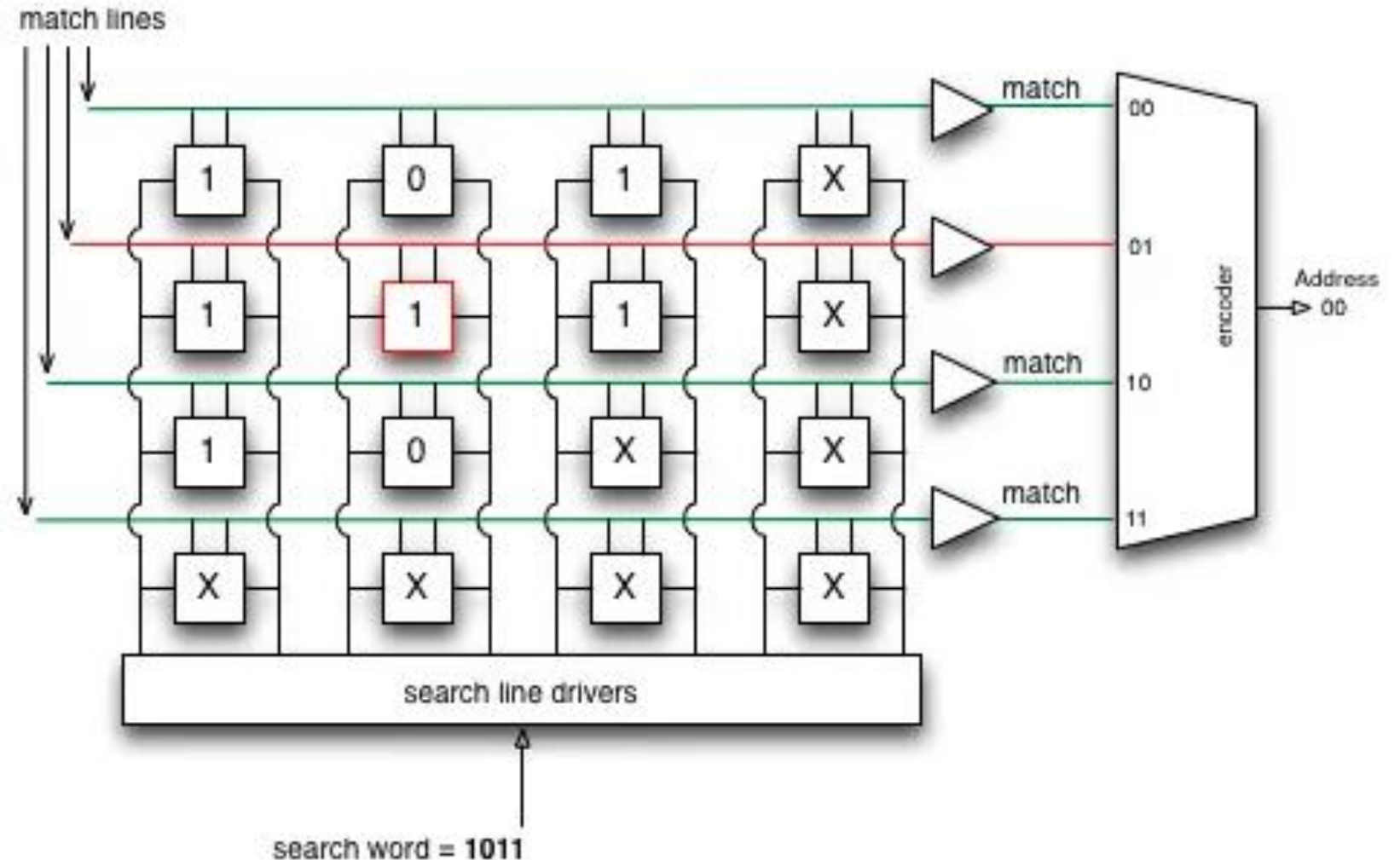
RIB Feeds FIB



Basic TCAM Architecture

Image Source: <http://thenetworksherpa.com/tcam-in-the-forwarding-engine/>

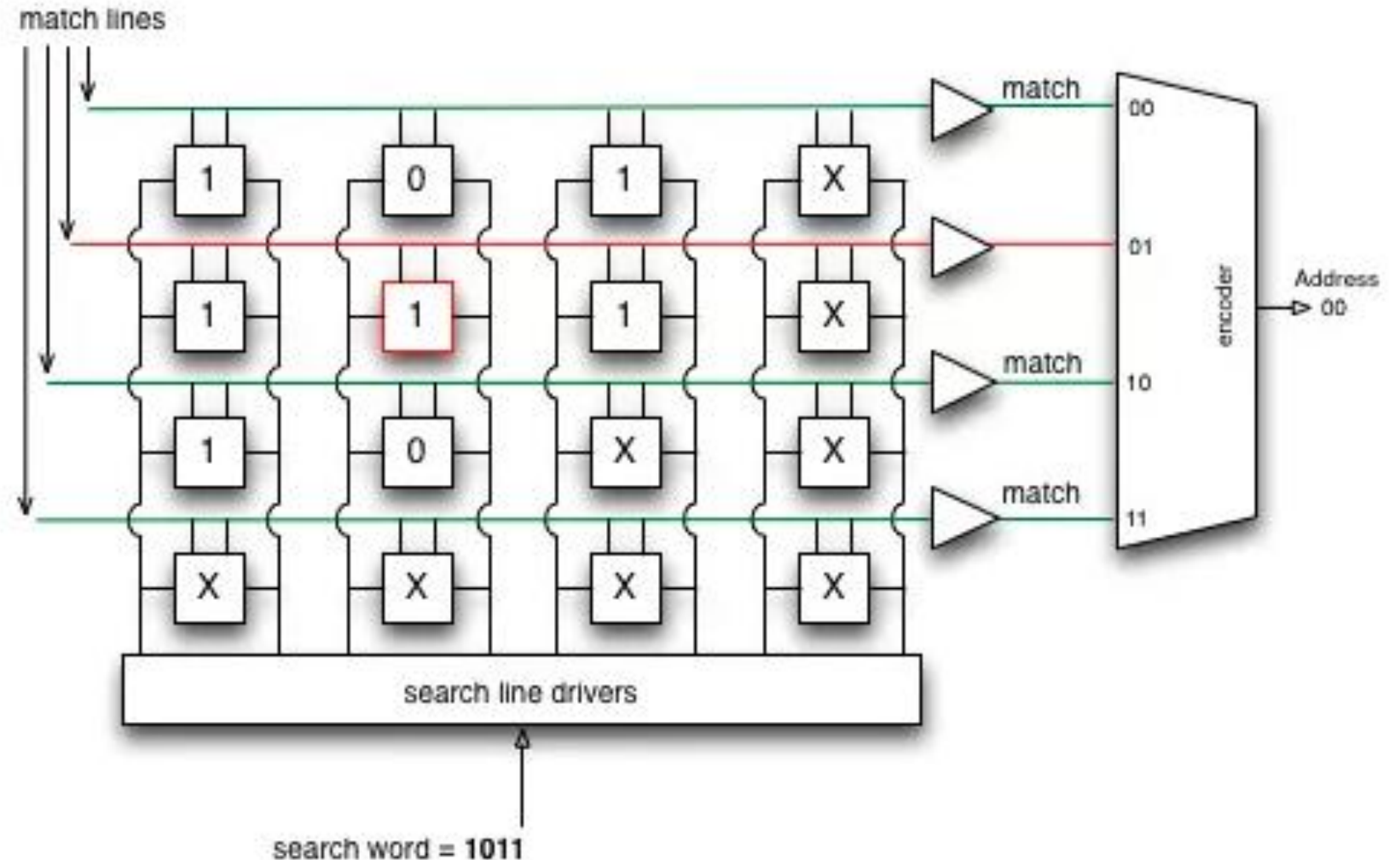
Pfx/mask	TCAM format
101/3	101X
111/3	111X
10/2	10XX
0/0	XXXX



Basic TCAM Architecture

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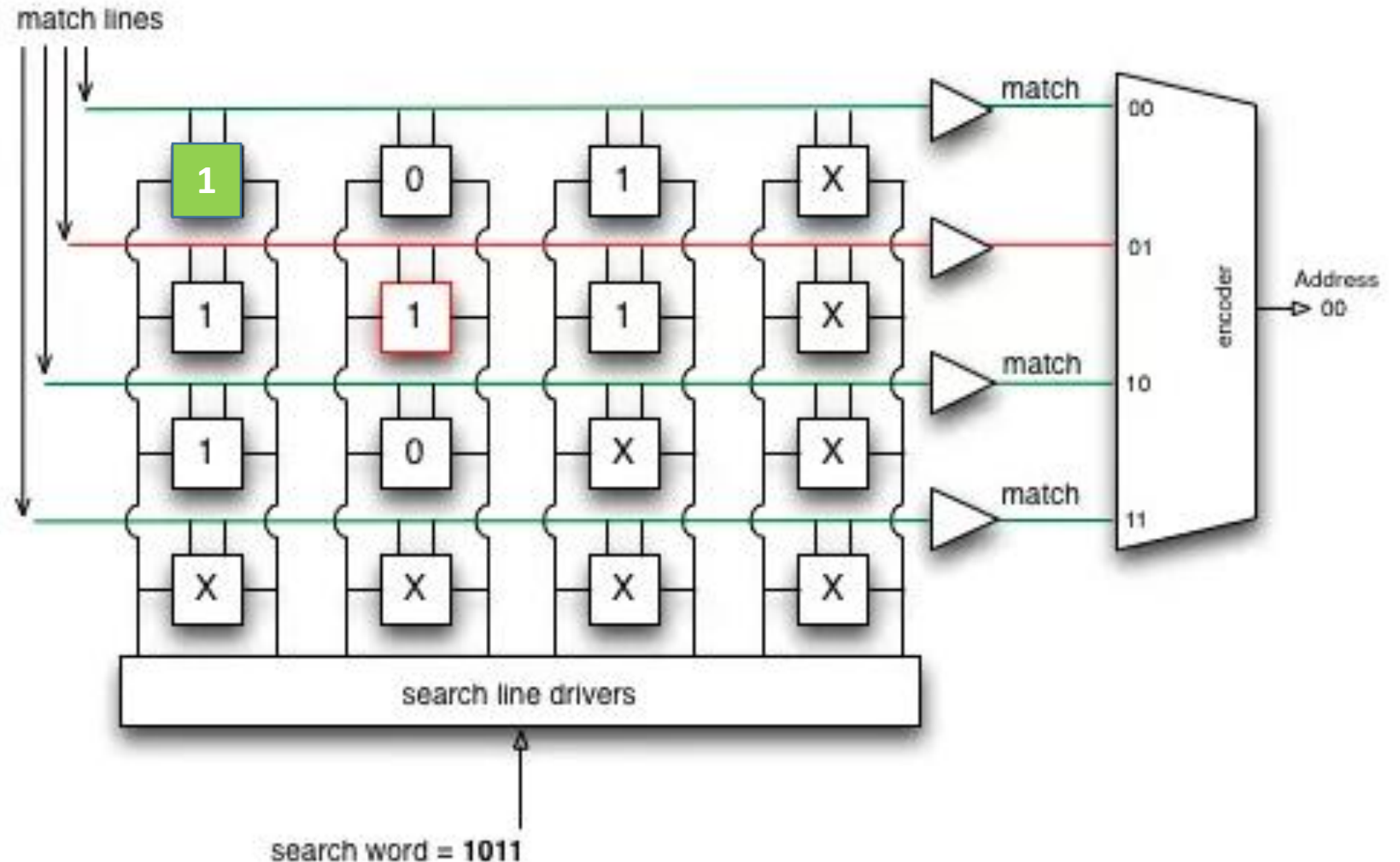


Let us try to match 1010

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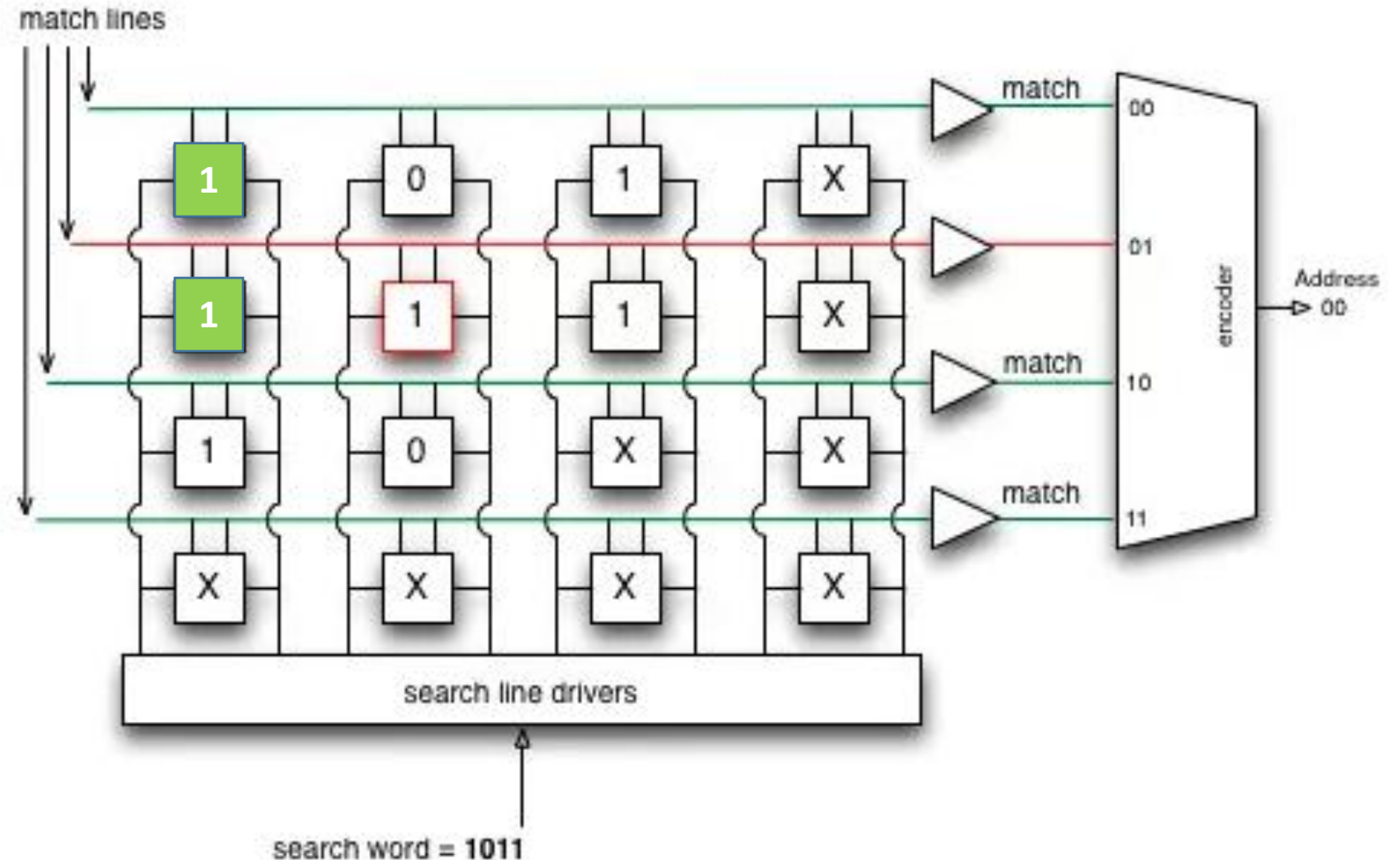


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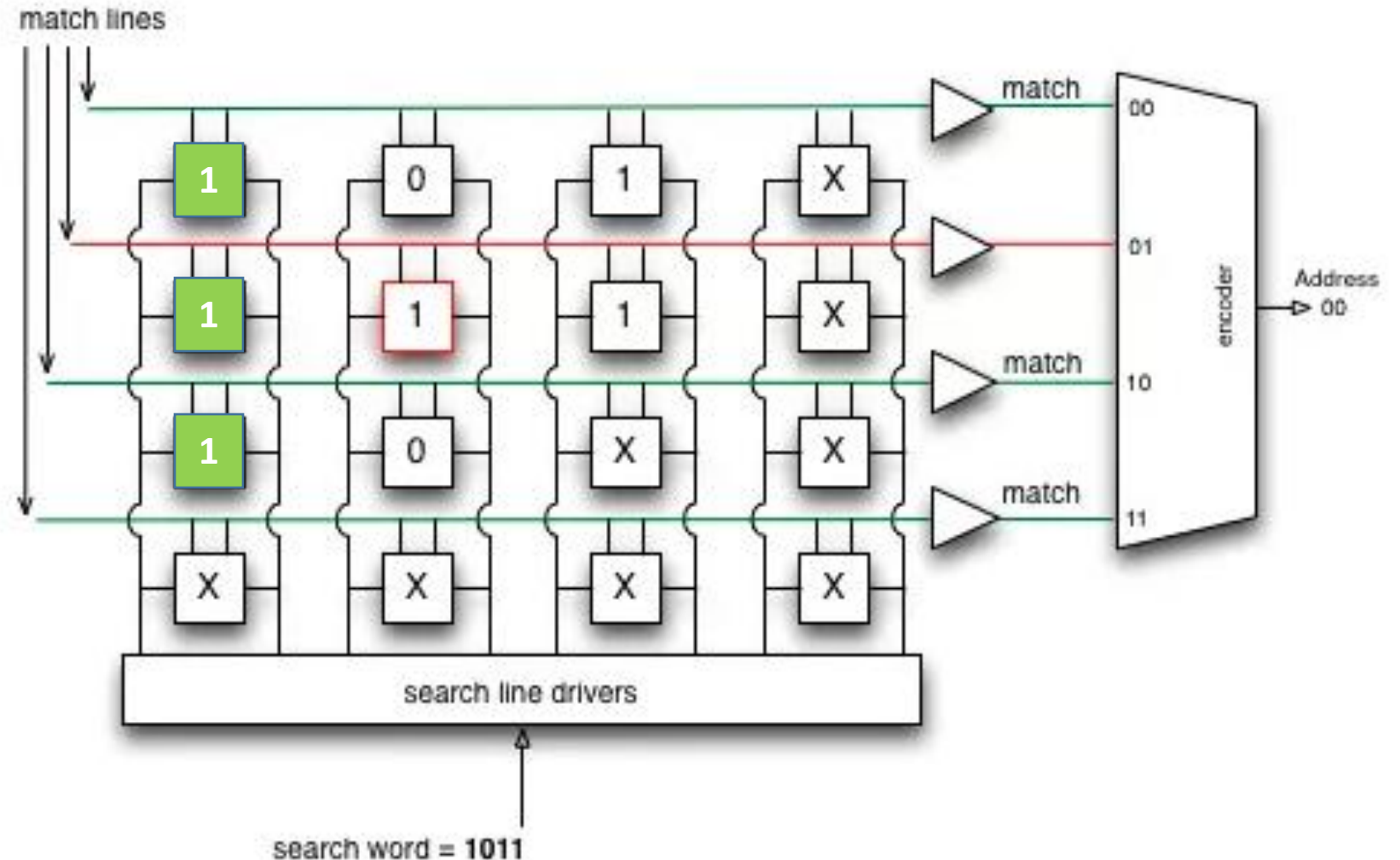


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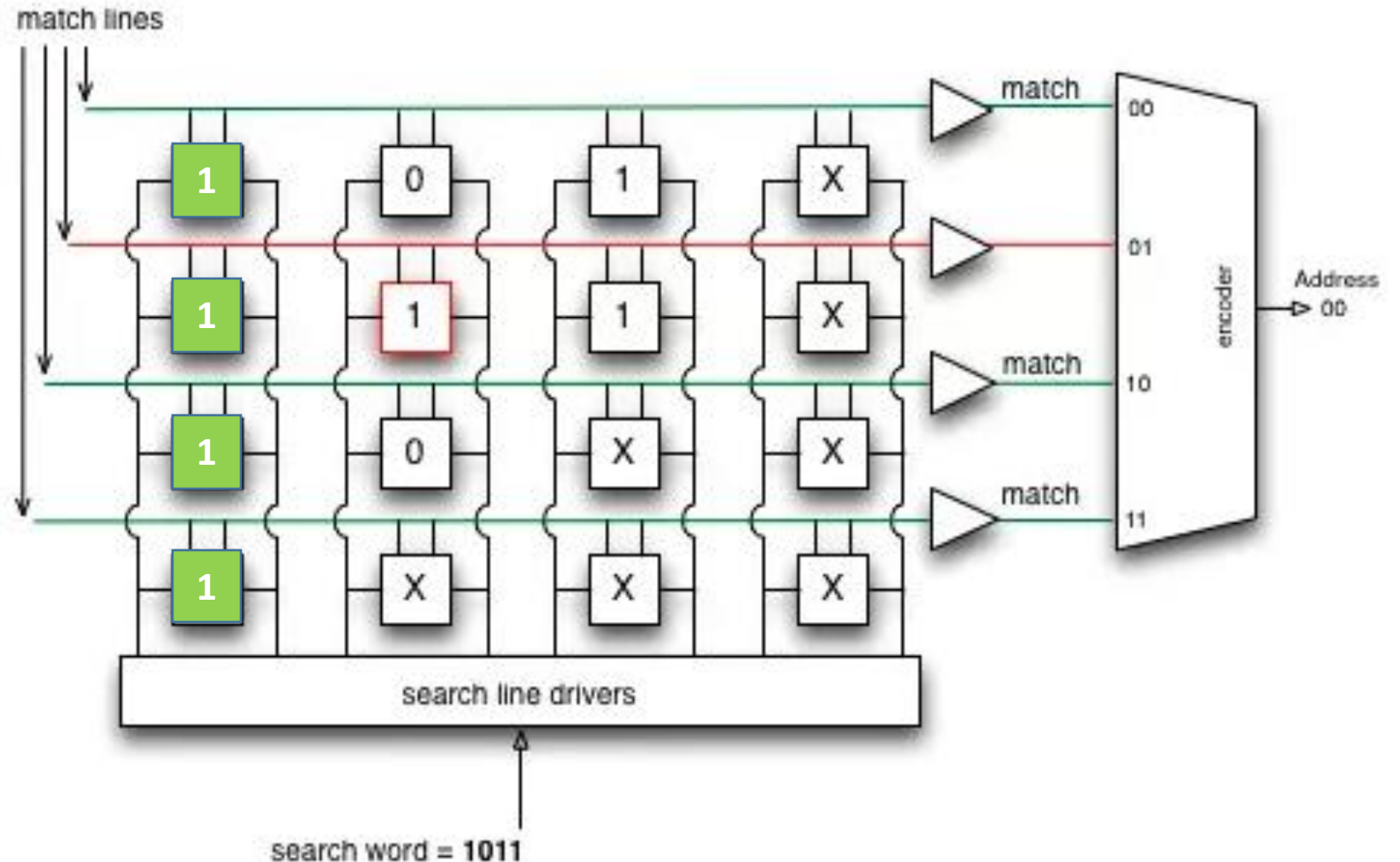


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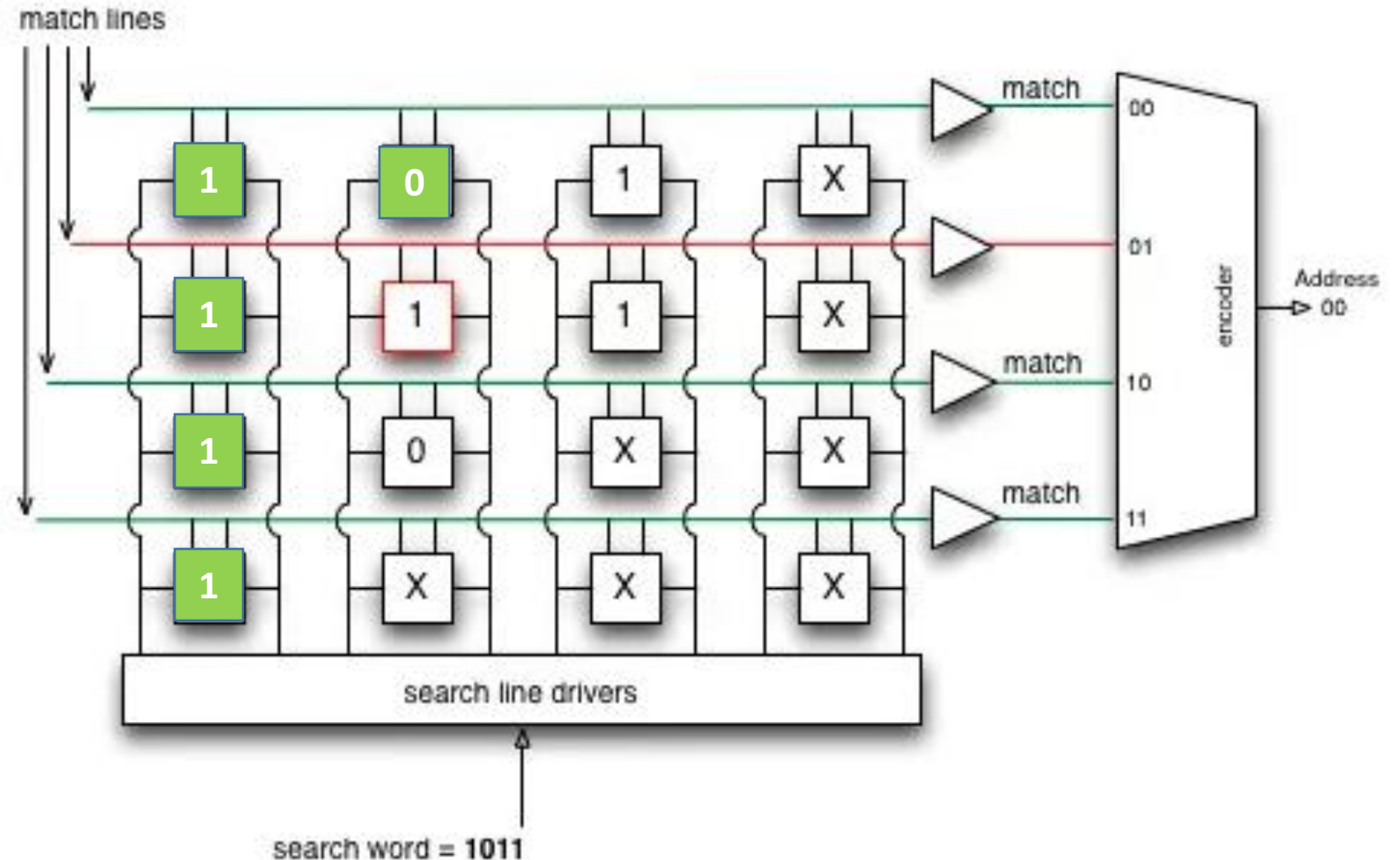


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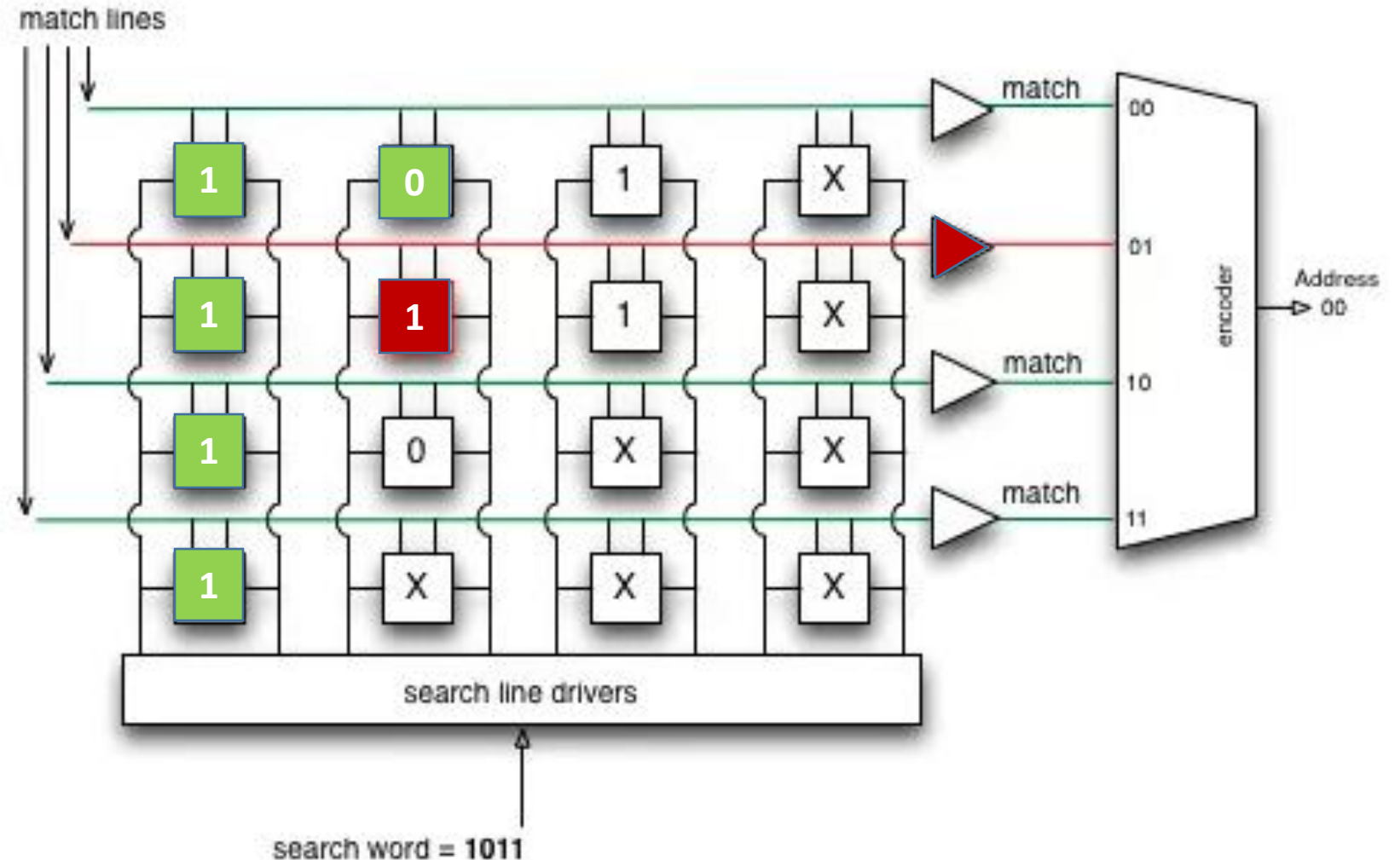


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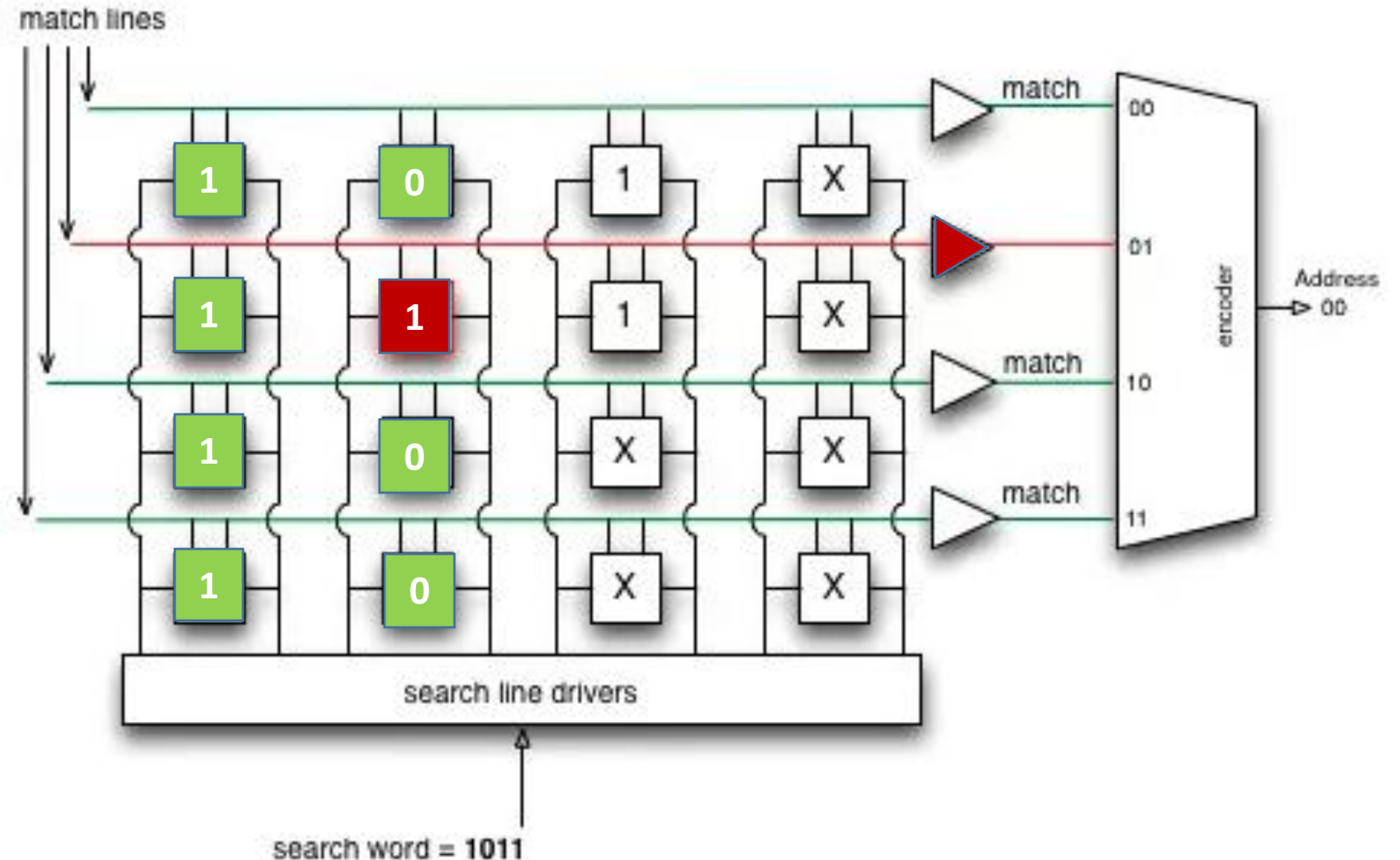


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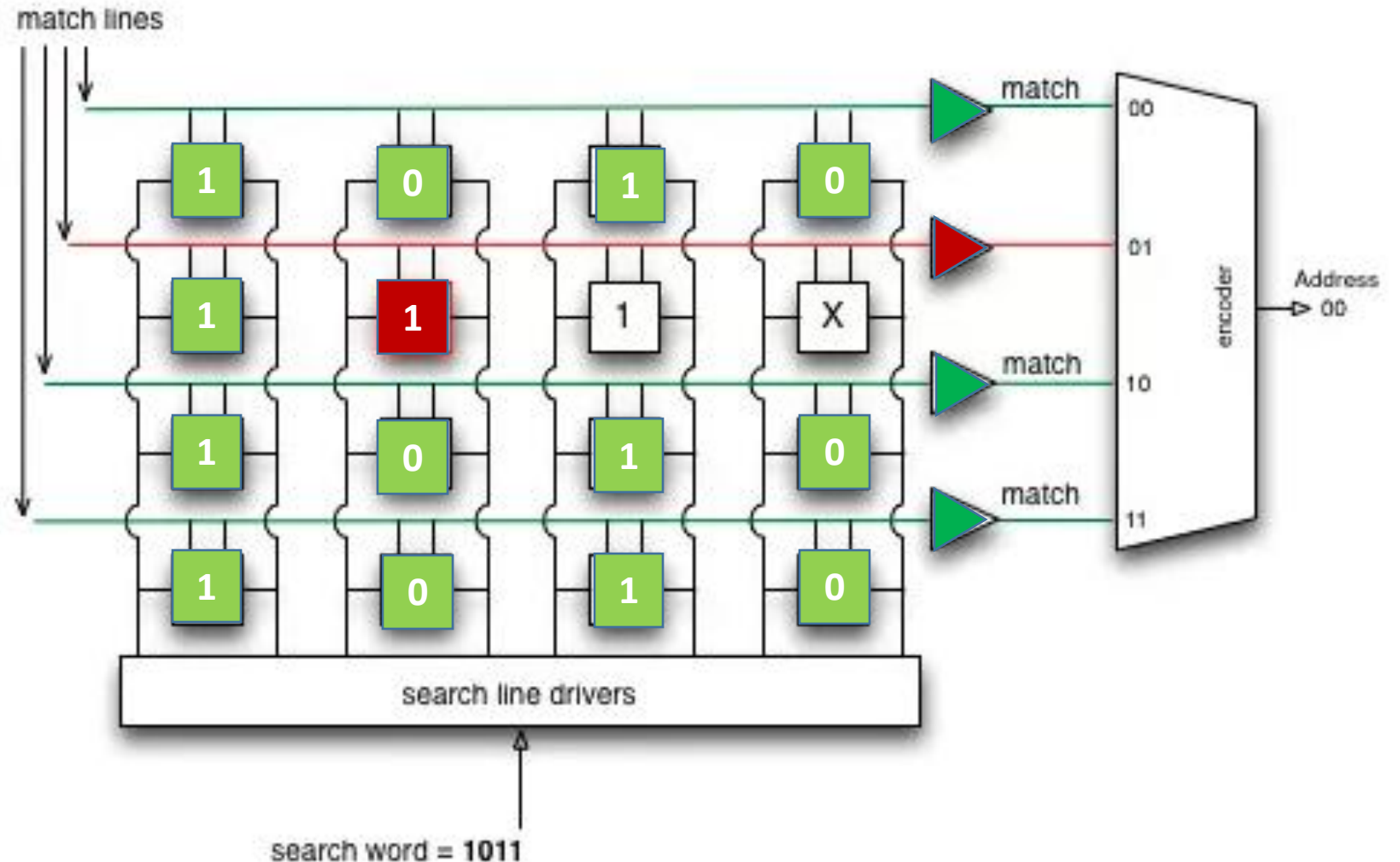


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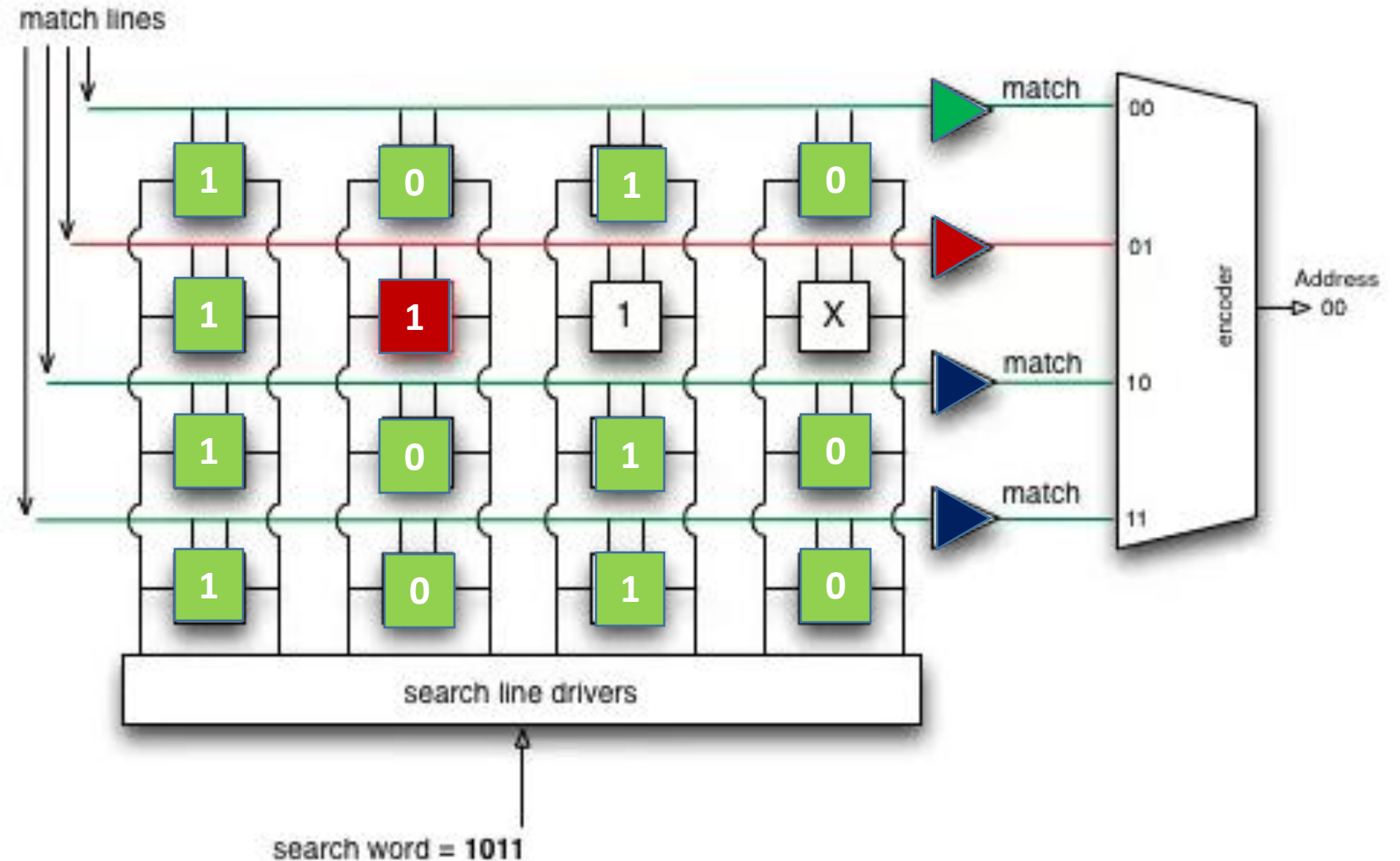


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Pfx/mask	TCAM format
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10/2	10XX
0/0	XXXX



Let us try to match 1010

Software Defined Networking

Software-defined networking (SDN) is a network framework which involves in **separating** a network's **control functions** from its **data forwarding functions**, centralizing its intelligence, and abstracting its underlying architecture from applications and services.

Control and Data Plane

- **Control plane**
 - The module which takes all decisions, basically an instructor
 - The routing algorithm
- **Data plane**
 - The module which carries out the tasks given by the control plane
 - Forwarding of Packets



Control and Data Plane

- Traditional networking devices are proprietary
 - Vendors decide software (control plane) and hardware (data plane)
- No standardization

***These two modules are "baked"
in
Unchangeable***



What does separating control and data plane mean?

But what if they were separate?

- Vendors only provide the hardware (data plane)
- We decide the control plane by writing custom logic – **the software**



What does separating control and data plane mean?

But what if they were separate?

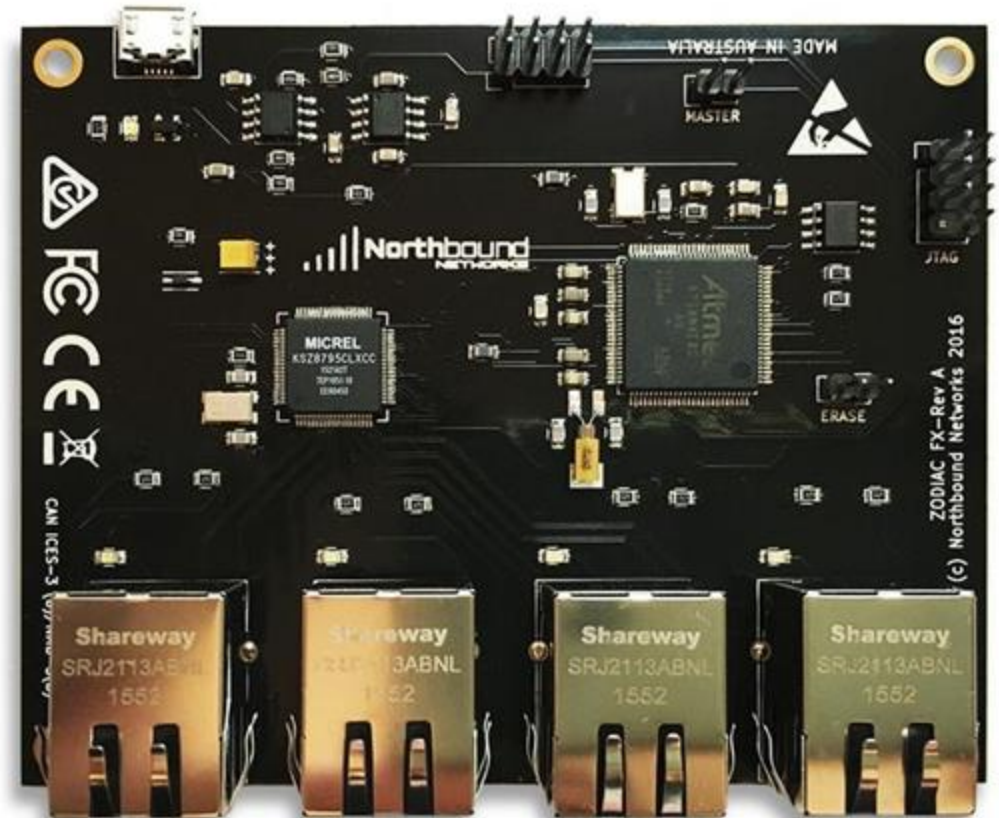
- Vendors only provide the hardware (data plane)
- We decide the control plane by writing custom logic

Advantages:

- Features are no longer limited to what the vendor provides
- Community development
- Longer life of products

How Does SDN Work

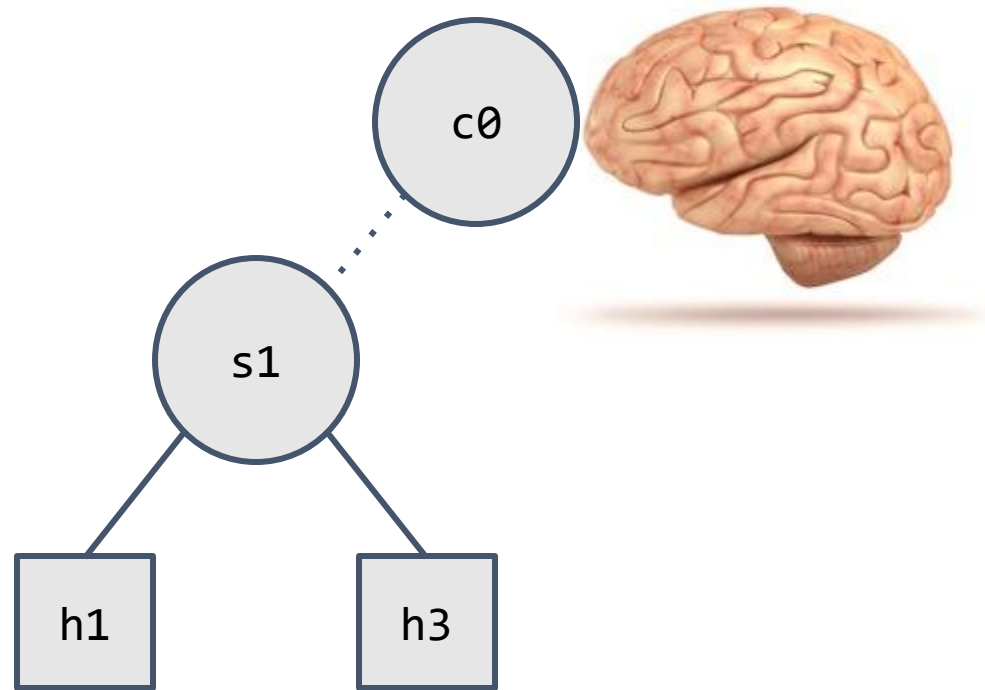
- Compared to traditional networks, a software defined network has 2 types of devices
 - Controller
 - Switches
- The switches in SDN are **blind**
 - No built-in features
 - Need to be instructed by the controller



Zodiac FX - A Tiny SDN Switch

SDN Controller

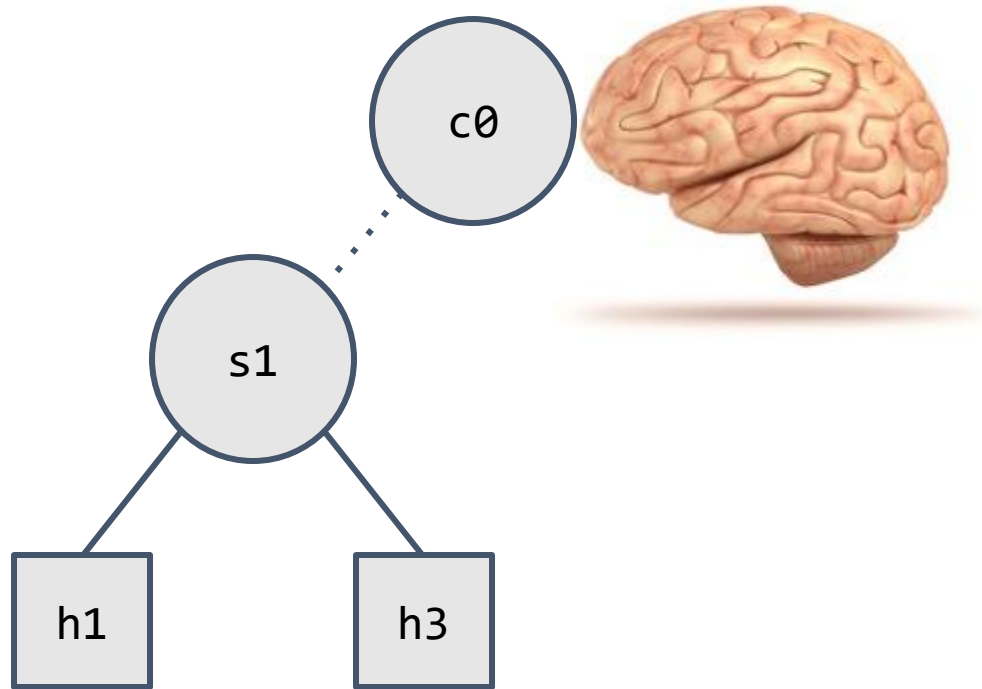
- “**Brains**” of the network
- All policies, routing logic are placed in the controller
- “**Teaches**” the switches what to do with an unknown packet



SDN Controller

- “Brains” of the network
- All policies, routing logic are placed in the controller
- “Teaches” the switches what to do with an unknown packet

Let us now see a scenario where a packet wants to go from h1 to h3 in an SDN environment



Forwarding Request from h1 to s1

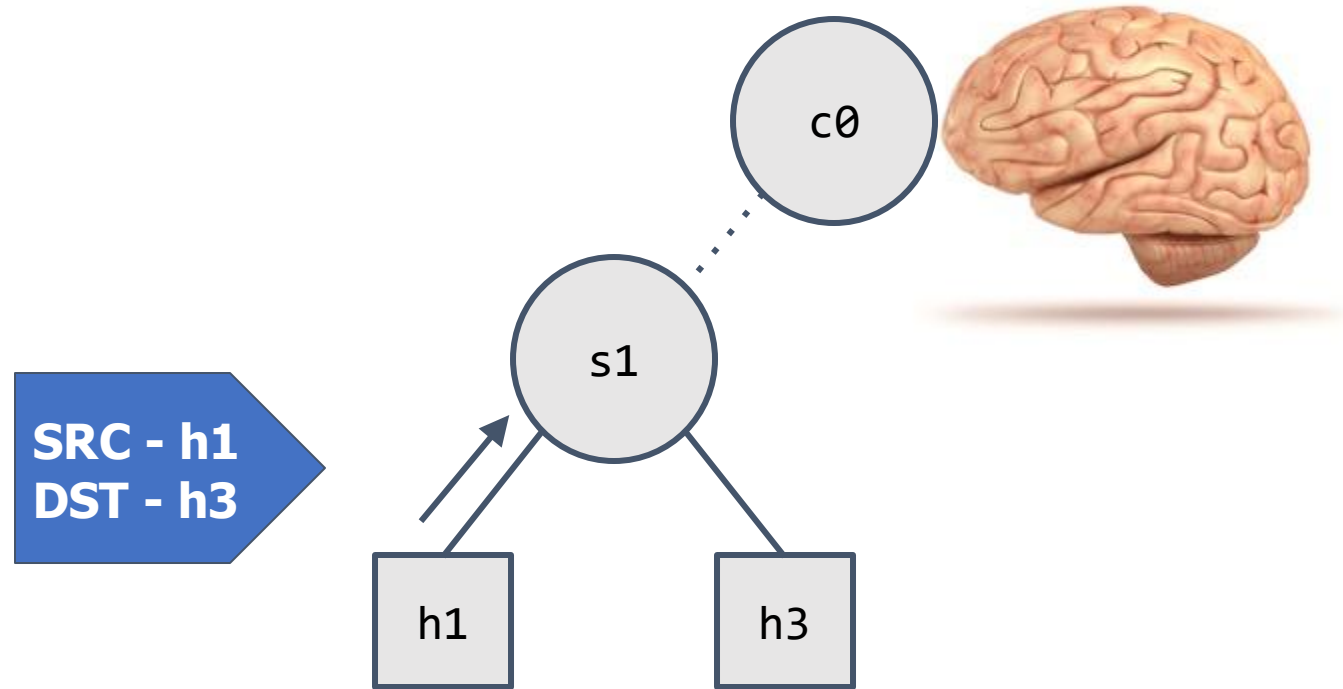
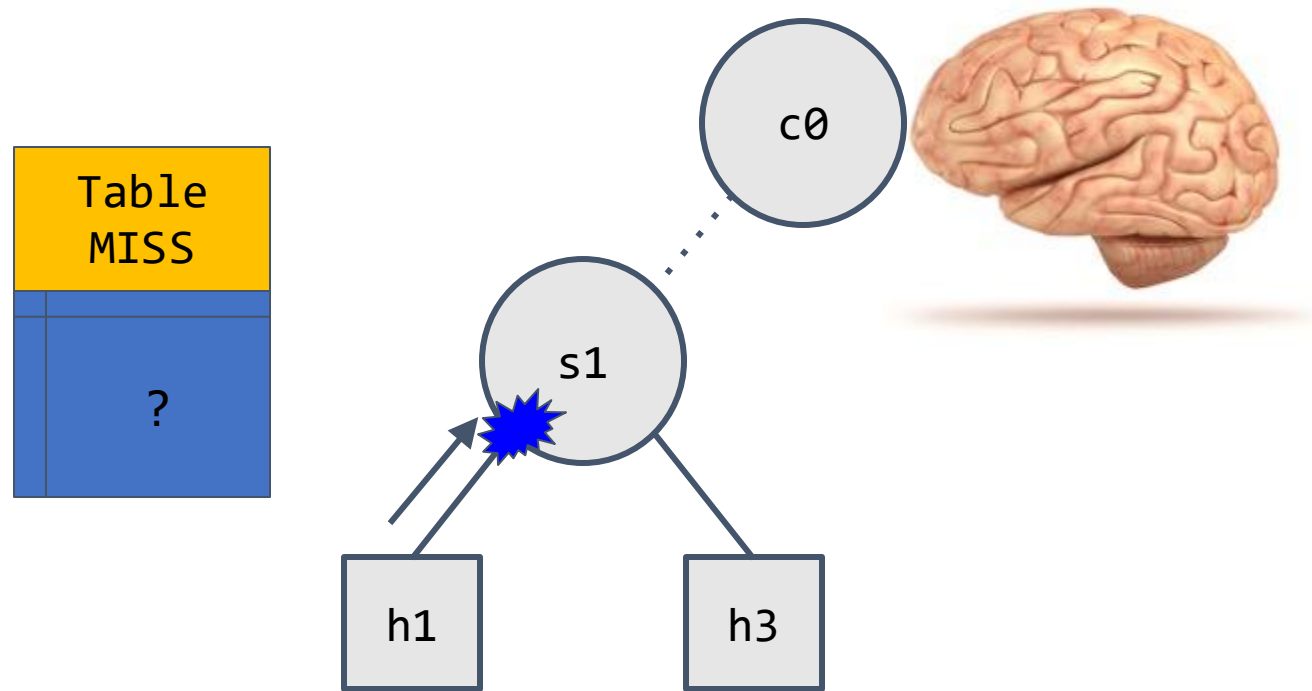


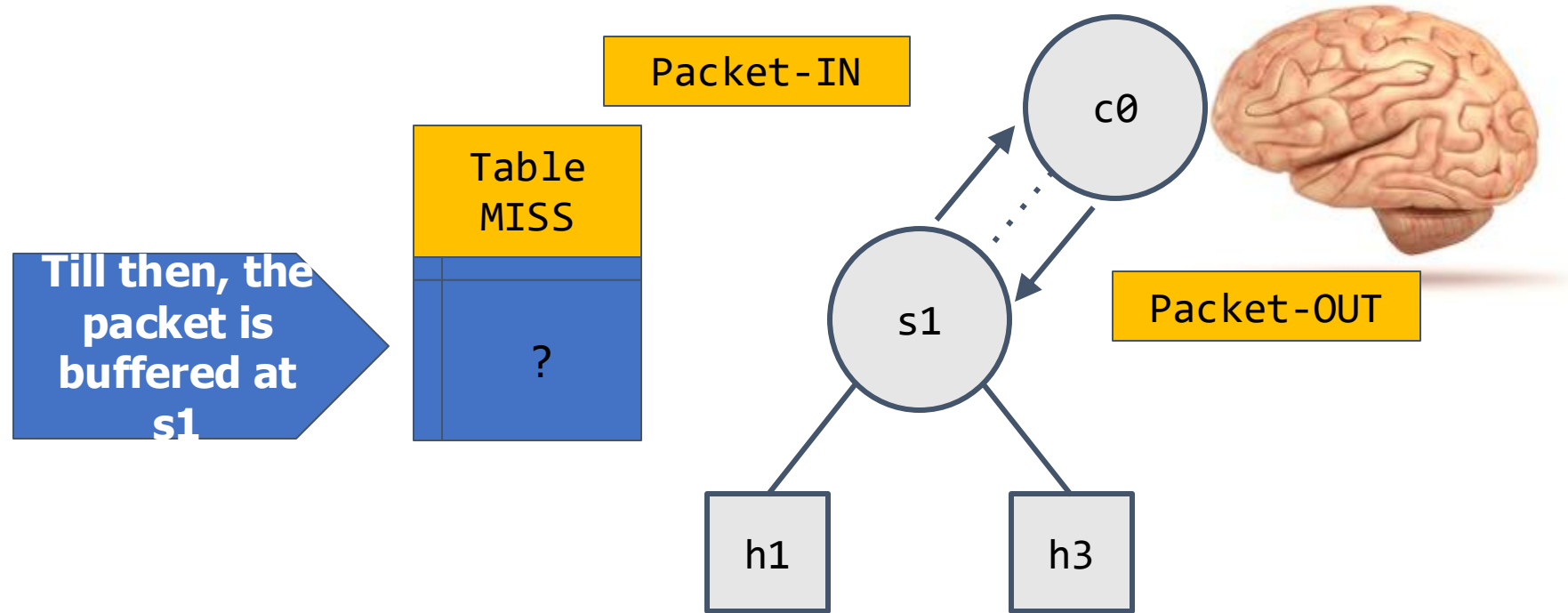
Table Miss at s1

No rule is there for $h1 \rightarrow h3$ at the beginning, so a FIB miss at S1



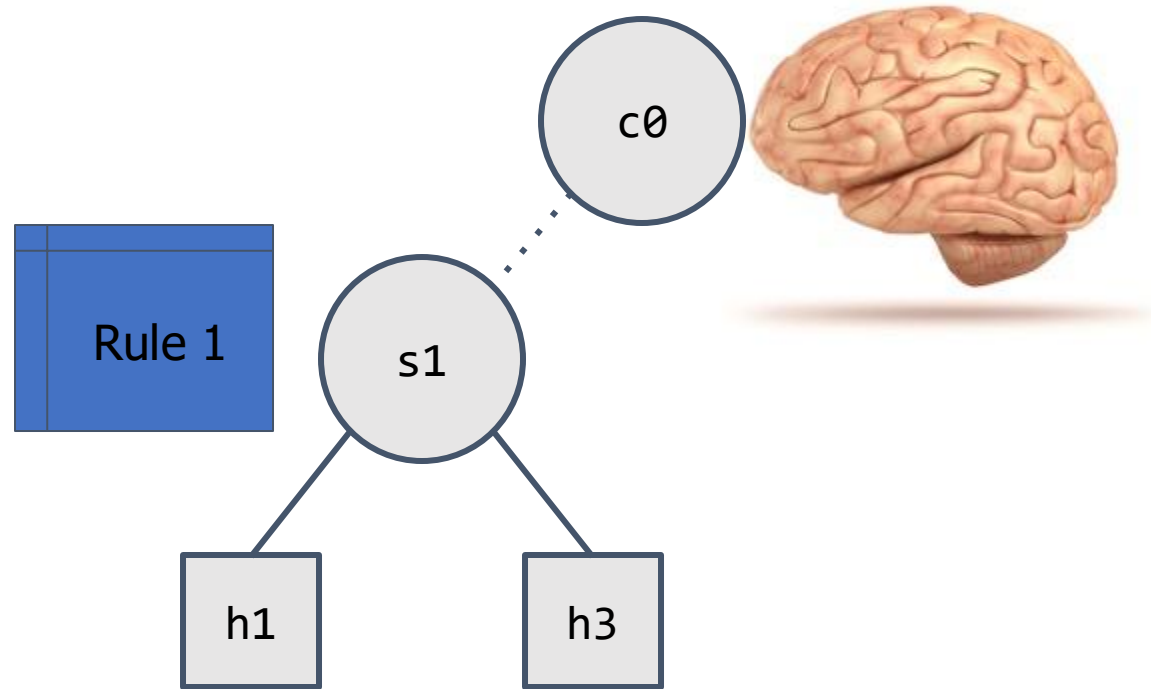
Packet-IN to Controller

The controller generates the rule based on a software program (the routing logic)



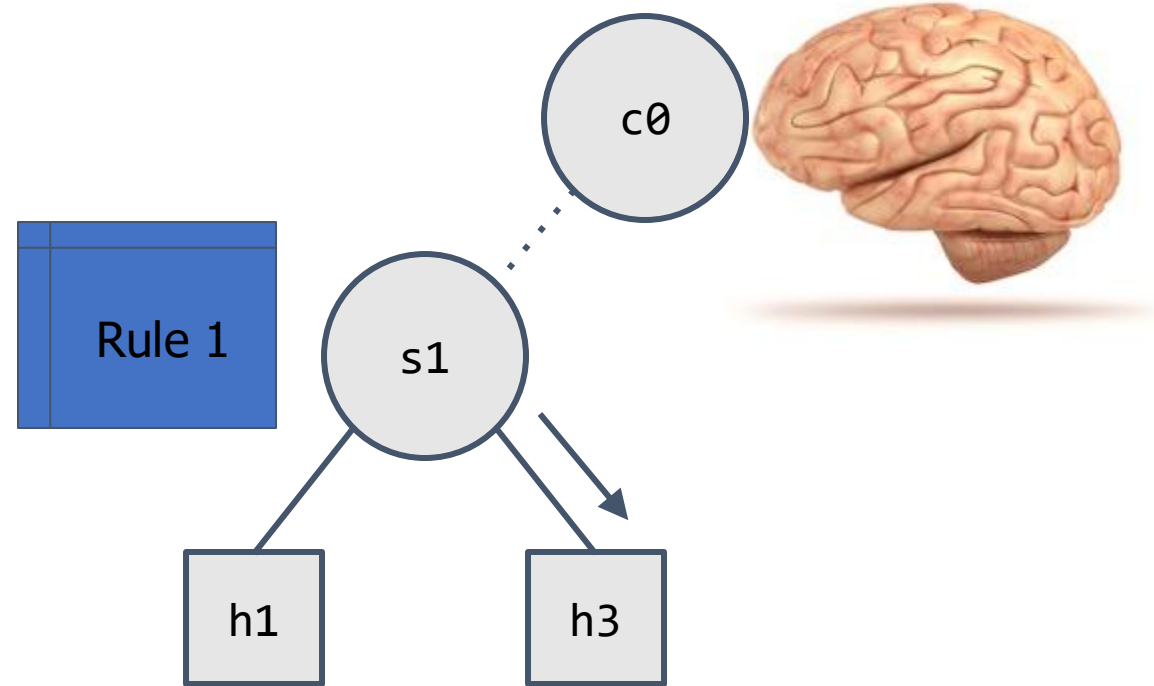
Flow Rule Set at s1...

The rule is installed at the FIB of S1



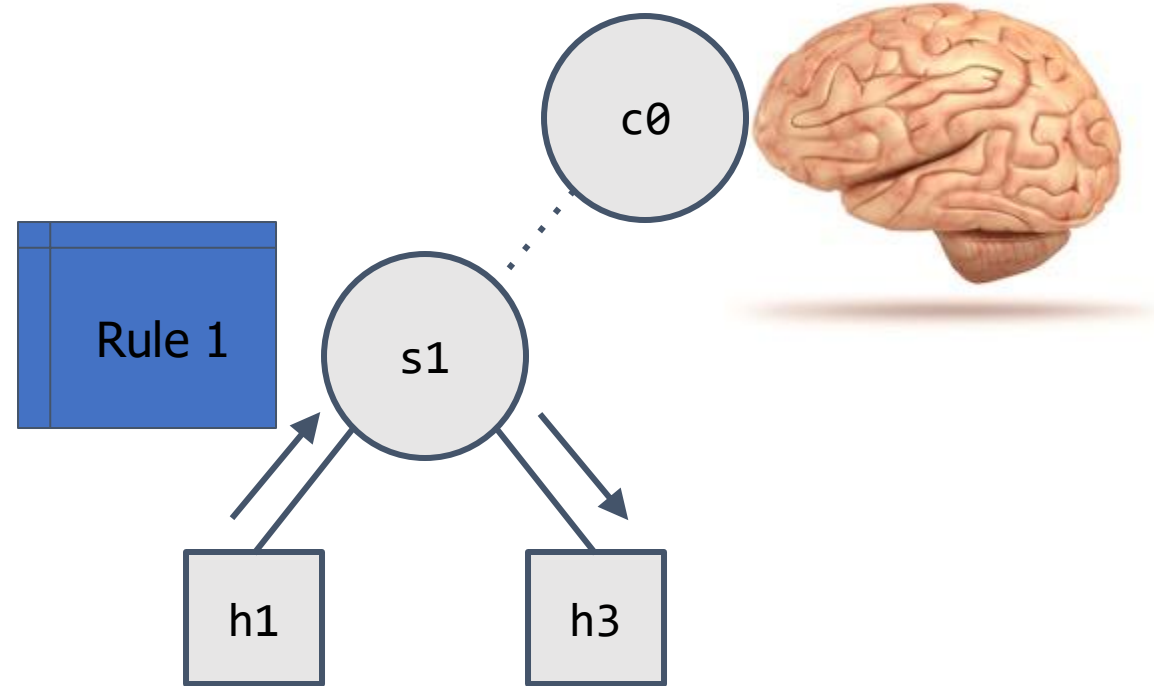
All Future Packets from h1->h3

The buffered packet is forwarded based on that rule



Buffered Packet Forwarded to h3...

All the future packets from h1->h3 follow the same rule



The SDN Architecture

Application layer

Applications,
running on physical
or virtual hosts



Control layer

Network controller



Northbound
APIs

Infrastructure layer

Programmable
switches



Southbound
API

Image source: qmonnet.github.io

Traditional Network vs SDN

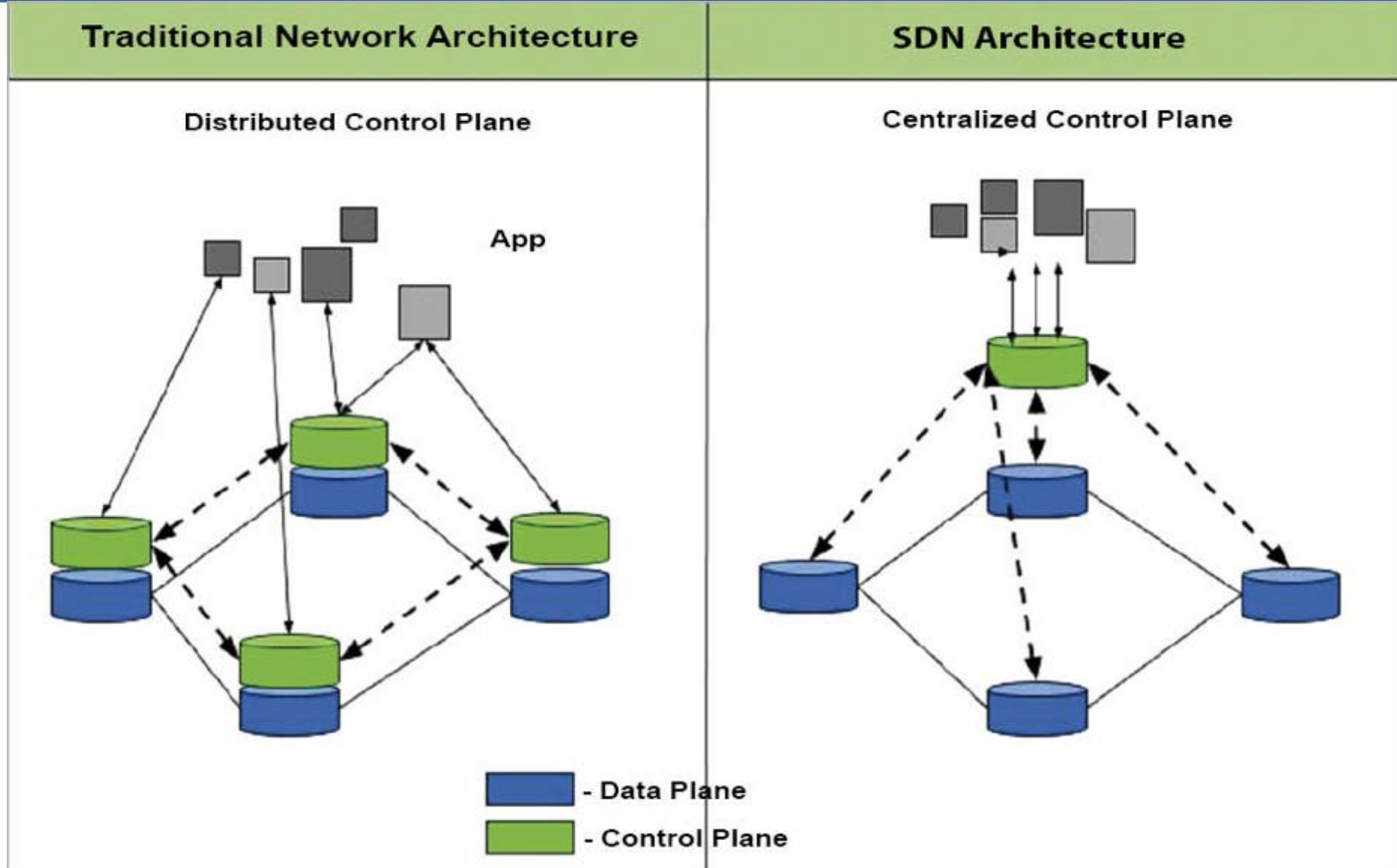


Image source: <http://opensourceforu.com/>